

USING CAPE TO ESTIMATE FUTURE STOCK RETURNS

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Recent Wall Street research suggests the prospect of relatively low stock returns over the next decade (*e.g.*, 1%/y real return).¹ This is a significant reduction compared to the prior decade's realized real return of 10.5%/y. The prospect of low future returns stems from high stock market valuation as measured by CAPE (Shiller's S&P 500 cyclically adjusted P/E). Historically, high levels of CAPE are associated with relatively low long-term future stock returns.

However, using CAPE to generate future stock return estimates can produce significantly different forecasts depending on how CAPE is specified in the forecast model and whether there are other explanatory variables. While CAPE is viewed as an important predictor of future stock returns, there is little consensus on how to use CAPE for future return estimation.

We compare various CAPE models and identify that $(1/\text{CAPE} - R)$ (*i.e.*, stock earnings yield – 10y US Treasury real yield) – also known as “stock excess yield” or the “Fed Model” – has delivered the best estimation performance (of those models considered), regardless of the level of CAPE.²

CIO Takeaways

1. $(1/\text{CAPE} - R)$ (*i.e.*, stock earnings yield – 10y US Treasury real yield) has been – both theoretically and empirically – the best way to incorporate CAPE in estimating future 10y stock returns, especially when CAPE is high.
2. Historically, the recent reading of $(1/\text{CAPE} - R)$ suggests a 4.6%/y real return for stocks over the next decade (*vs.* 10.5%/y over the past decade), translating to a 6.6%-7.6%/y realized nominal return assuming 2%-3%/y inflation.

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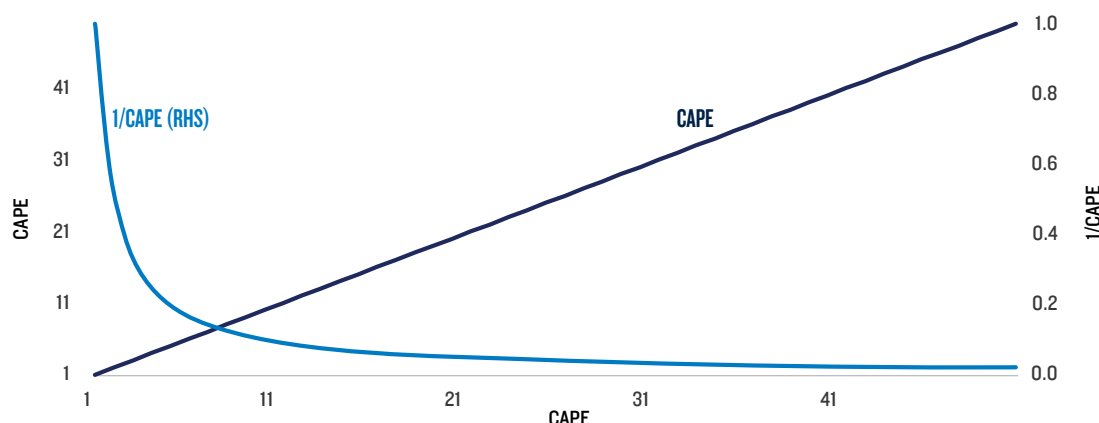
Why Does CAPE Specification Matter?

Many investors use CAPE to estimate future (say, over the next 10y) stock returns with a linear regression model:³

$$E(r_{t,t+10}) = \alpha_t + \beta_t \times "CAPE_t" + \varepsilon_{t,t+10} \quad (1)$$

Investors employ various specifications for “CAPE” such as CAPE (used by Barclays),⁴ $\ln(\text{CAPE})$ (used by Goldman Sachs),⁵ and $1/\text{CAPE}$ (used by PGIM Multi-Asset Solutions).⁶ How much does the specification of “CAPE” matter? After all, “CAPE” is “CAPE.” However, different versions of CAPE differ significantly in terms of magnitude and dynamics, so the best linear fit to stock returns will be sensitive to the specification of CAPE. For example, consider CAPE and $1/\text{CAPE}$. While CAPE is (of course) linear in CAPE, $1/\text{CAPE}$ changes rapidly when CAPE is low and changes little when CAPE is large (Figure 1).

Figure 1: CAPE and 1/CAPE

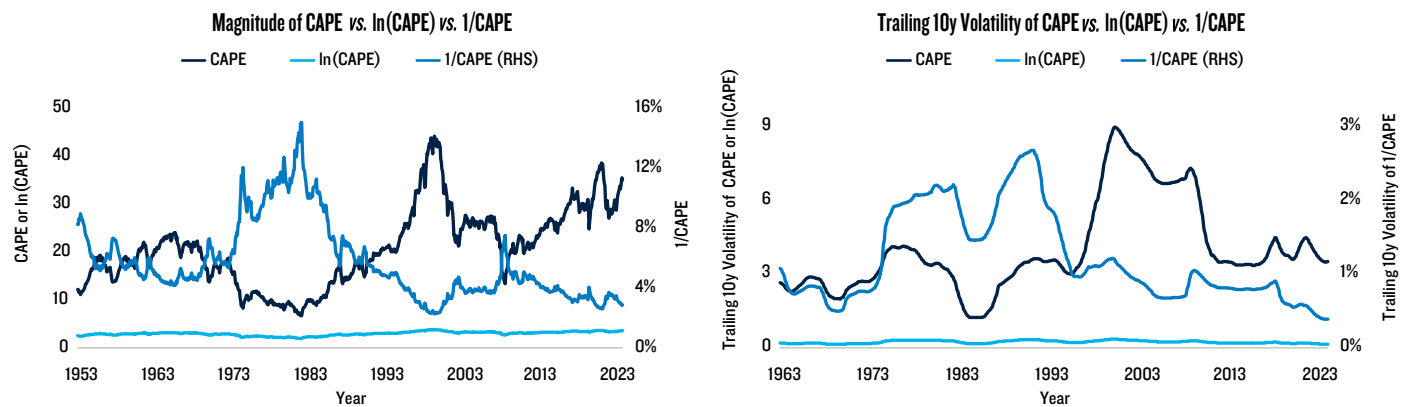


Source: PMA. Provided for illustrative purposes only.

In fact, Figure 2 shows that different CAPE specifications have indeed differed wildly, both in terms of magnitude and volatility.

- 1 See *Updating Our Long-Term Return Forecast for US Equities to Incorporate the Current High Level of Market Concentration*, Goldman Sachs, October 2024. All returns in our paper, either nominal or real (CPI adjusted), are total annualized returns.
- 2 CIOs may also consider enhancing estimation accuracy by adding supplementary variables – e.g., Goldman Sachs (2024) incorporates other variables (in addition to the logs of CAPE and 10y US Treasury nominal yield) including stock market concentration, economic contraction frequency, and corporate profitability. However, the incremental benefit might be insignificant given the forecast record of $(1/\text{CAPE} - R)$. We choose to focus on CAPE and R – both are widely used, important predictors of future stock returns – instead of exploring other explanatory variables. See *Higher Bond Yields & the Fed Model – Implications for Future Stock-Bond Relative Returns*, PGIM Multi-Asset Solutions, November 2023 regarding our adjusted version of the Fed Model. A key criticism of the Fed Model is its comparison of a *real* number (S&P 500 earnings yield, inverse of the 12m forward P/E ratio, which is a real quantity as the impacts of inflation on price and on earnings offset each other) to a *nominal* one (nominal 10y Treasury yield, Y). Consequently, we define the Fed Model using the real (i.e., inflation-adjusted) 10y Treasury yield. Additionally, we replace the 12m forward P/E ratio with CAPE as it provides a longer history for our study. CAPE takes a 10y average of inflation-adjusted earnings to smooth out volatility of earnings over different periods of a business cycle. See C. Asness (2003), “Fight the Fed Model,” *The Journal of Portfolio Management*, 30 (1) 11-24; J. Estrada (2009), “The Fed Model: The Bad, the Worse, and the Ugly,” *The Quarterly Review of Economics and Finance*, 49 (2) 214-238; and J. Y. Campbell and R. J. Shiller (1988), “Stock Prices, Earnings and Expected Dividends,” *The Journal of Finance*, 43 (3) 661-676.
- 3 We focus on *real* total returns since CAPE, a price-to-earnings ratio, is a *real* quantity and should be used to estimate *real* instead of nominal returns. See C. Asness (2003), “Fight the Fed Model,” *The Journal of Portfolio Management*, 30 (1) 11-24.
- 4 See *Time for More Defensive Equity Positioning*, Barclays, November 2023.
- 5 See *Updating Our Long-Term Return Forecast for US Equities to Incorporate the Current High Level of Market Concentration*, Goldman Sachs, October 2024.
- 6 See *Forecasting Long-Term Equity Returns: A Comparison of Popular Methodologies*, PMA, September 2018.

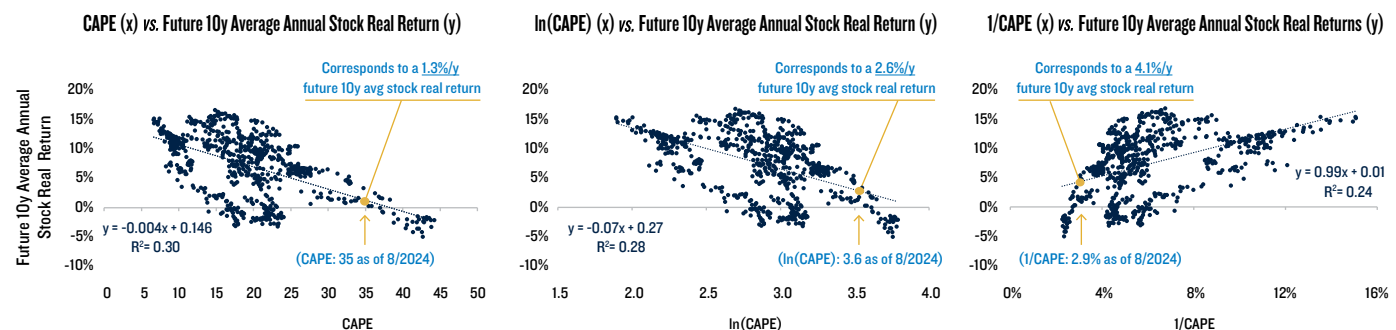
Figure 2: Magnitude & Volatility of Various CAPE Specifications; Monthly, 4/1953 – 8/2024



Note: The magnitude and volatility of $\ln(\text{CAPE})$, although less legible due to scale, follow the same trend as those of CAPE since they are positively related. Source: Robert Shiller and PMA. Provided for illustrative purposes only.

So, not surprisingly, different CAPE specifications exhibit different relationships with future stock returns – especially when CAPE is relatively high – leading to divergent return estimates (Figure 3). For instance, the recent values of CAPE, $\ln(\text{CAPE})$ and $1/\text{CAPE}$ (as of August 2024) correspond to future 10y stock real return estimates of 1.3%/y, 2.6%/y and 4.1%/y, respectively, showing a 280bp/y spread over 10y!

Figure 3: CAPE vs. $\ln(\text{CAPE})$ vs. $1/\text{CAPE}$: Relationship with Future 10y Average Annual Stock Real Returns; 4/1953 – 9/2014



Note: Calculations are based on real total returns. Source: Robert Shiller and PMA. Provided for illustrative purposes only.

Which CAPE Model Specification Has Performed Best?

Theoretically, stock returns would be expected to have a linear relationship with $1/\text{CAPE}$ – not CAPE or $\ln(\text{CAPE})$ – suggesting that $1/\text{CAPE}$ might be the best empirical specification for “CAPE.”⁷ However, we “let the data speak” and compare the **evolution of estimation accuracy** across different CAPE models **with an expanding window** (Figures 4 & 5; see Appendix for model details).⁸

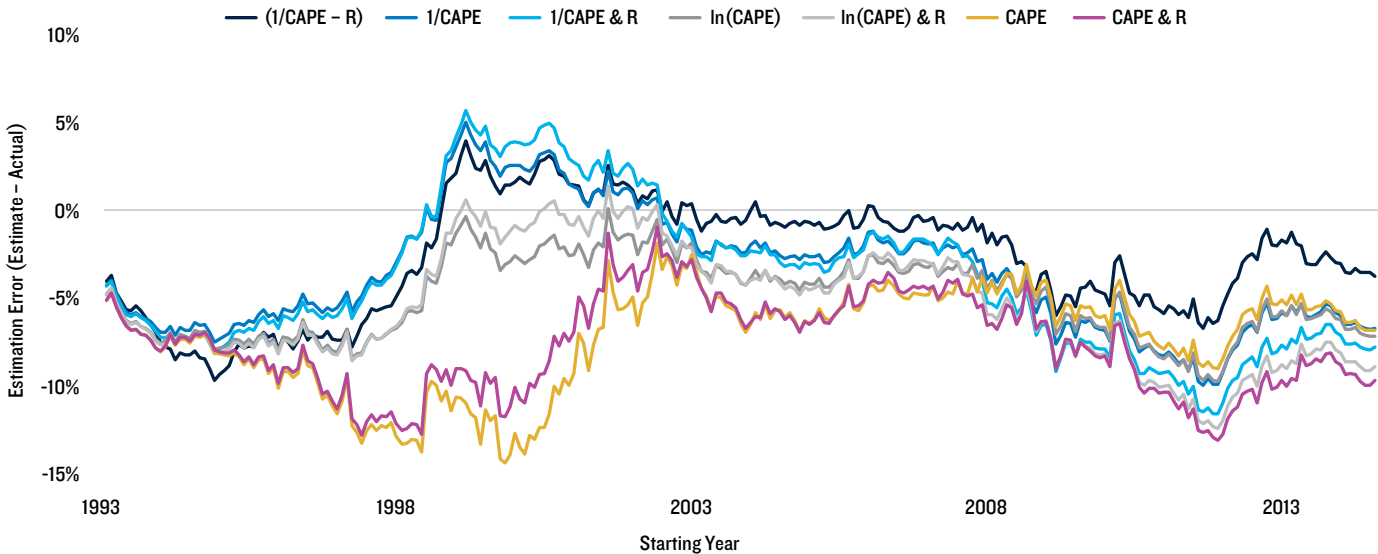
We also consider adding 10y US Treasury *real* yields (“R”) as an explanatory variable, along with CAPE, to account for the impact of *real* interest rates on future stock *real* returns. High CAPE levels (*i.e.*, low equity earnings yields, $1/\text{CAPE}$) may not signal valuations that are too high, with concomitant low future returns, if real bond yields are low as well. What might matter more for forecasting real stock returns is the real yield difference between stocks and bonds. Consequently, the difference, $(1/\text{CAPE} - R)$, aka “stock excess yield” or the “Fed Model,” is expected to be *positively* related to future stock returns. When $(1/\text{CAPE} - R)$ increases (*i.e.*, stock yields increase relative to bond yields), stocks tend to be relatively undervalued and may offer higher future returns, *ceteris paribus*, to bring their earnings yield more in line with bonds.

7 See T. Philips & C. Ural (2016), “Uncloaking Campbell and Shiller’s CAPE: A Comprehensive Guide to Its Construction and Use,” *The Journal of Portfolio Management*, 43 (1) 109-125.

8 Note that we choose not to focus on goodness-of-fit (R^2) or statistical significance (t-statistic) because 1) they do not directly measure estimation performance as estimation error does, and 2) they are unreliable due to overlapping observations and endogeneity issues. See T. Philips and C. Ural (2016) on discussions on adjusted R^2 , adjusted t-statistic, and a more robust regression method (*i.e.*, the Theil-Sen estimator).

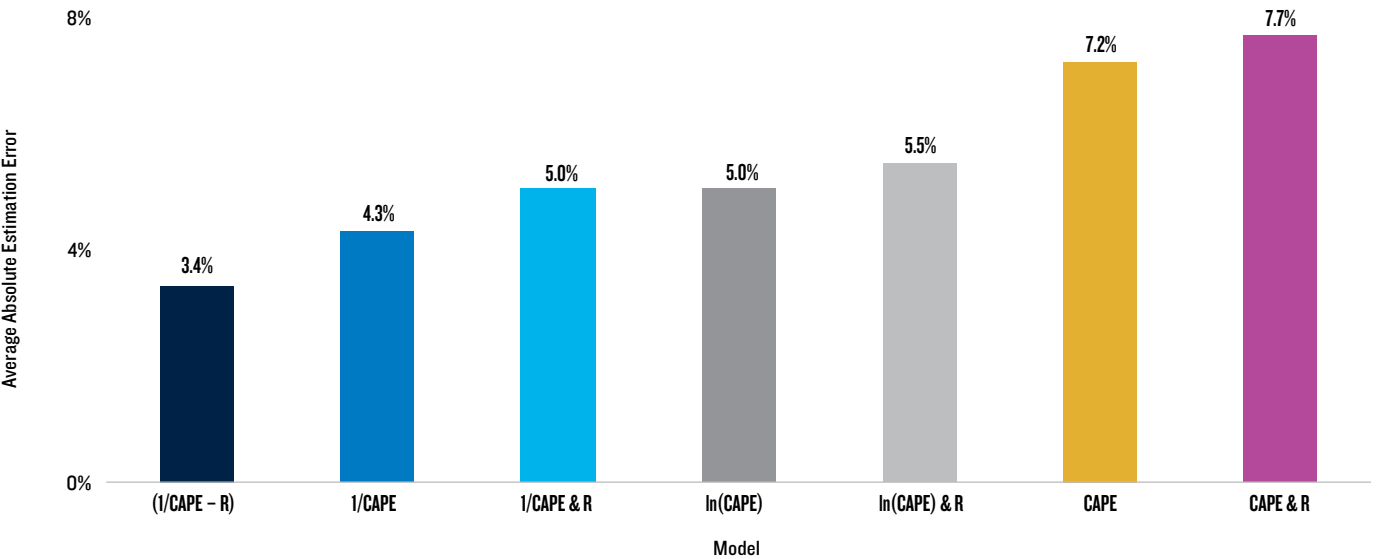
Overall, 1/CAPE-based models (with and without the additional regressor, R) have generally outperformed others with lower estimation errors over time, while CAPE-based models have persistently *underestimated* future stock returns. Furthermore, 1/CAPE-based models are intuitively easier to interpret, as stock earnings yield (1/CAPE) – a “return” – directly relates to stock returns. Among the models examined, (1/CAPE – R) has delivered the best forecast performance with the highest estimation accuracy (lowest average absolute estimation error). It has consistently topped other models since 2003.

Figure 4: Estimation Error (Estimate – Actual; Out of Sample; %/y) of Future 10y Average Annual Stock Real Returns Based on An Expanding Window; Monthly; 3/1993 – 9/2014



Note: The (1/CAPE – R) model is a special case of the 1/CAPE & R model where $\beta^R = -1 \times \beta^{(1/CAPE)}$. Calculations are based on real total returns. We run regressions and calculate estimation errors *every month* using data with an expanding window starting from April 1953 (when data become available) *until that month*, to avoid look-ahead bias. Regressions start in March 1993 (40y since April 1953) to allow for sufficient data to generate the first estimate. For 1/2003-9/2023, the real 10y yield is derived by subtracting the 10y breakeven inflation rate from the nominal 10y yield; for 1/1973-12/2002, we use the real 10y yield estimates from Barclays; for 4/1953-12/1972, the real 10y yield is approximated by subtracting the CPI inflation rate from the nominal 10y yield. All numbers (except for Barclays data) are averages of business days within a month. Source: Barclays, Datastream, Federal Reserve Bank of St. Louis, FRED, Haver Analytics, Robert Shiller, S&P, US Treasury and PMA. Provided for illustrative purposes only.

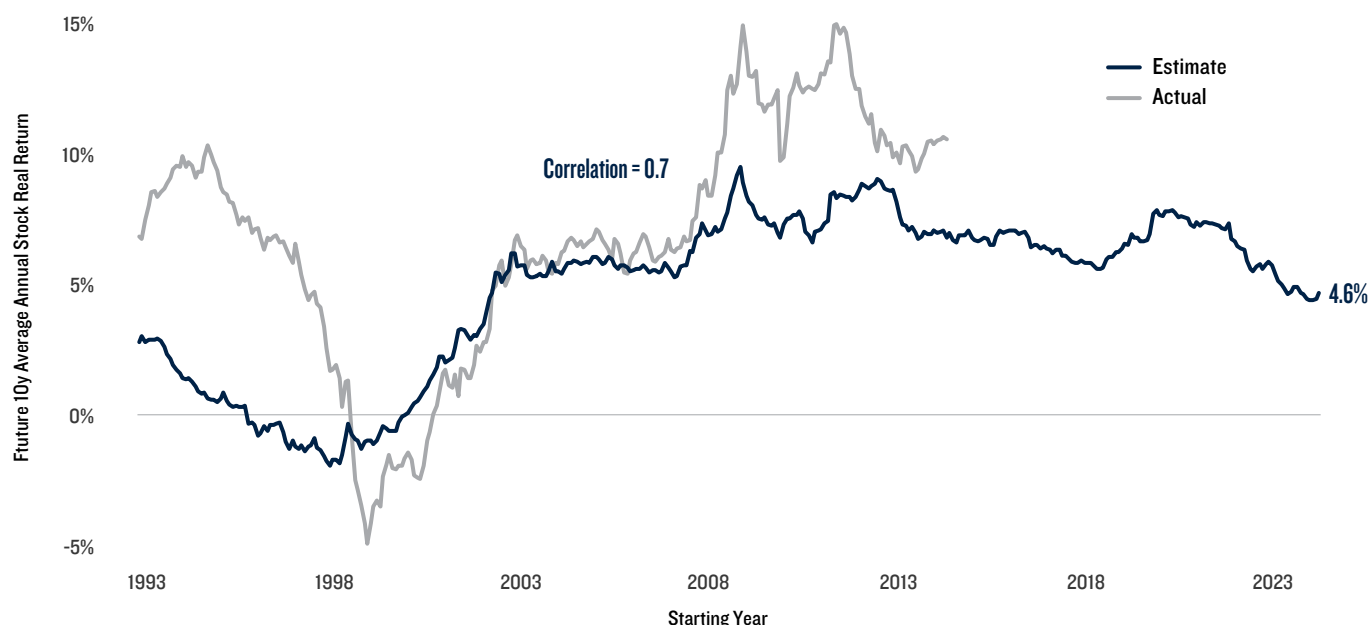
Figure 5: Average Absolute Estimation Error (%/y) of Future 10y Average Annual Stock Real Returns; 3/1993 – 9/2014



Same note and source as Figure 4. Provided for illustrative purposes only.

The estimates based on (1/CAPE – R) have been highly correlated with actual returns, as expected, with a correlation of 0.7 (Figure 6).

Figure 6: Future 10y Average Annual Stock Real Returns: Estimate (Using $(1/\text{CAPE} - R)$) vs. Actual; Monthly; 3/1993 – 8/2024



Same note and source as Figure 4. Regressions are run *every month* using data with an expanding window starting from April 1953 (when data become available) *until that month*, to avoid look-ahead bias. Regressions and estimates start in March 1993 (40y since April 1953) to include a sufficiently long history for robustness. Provided for illustrative purposes only.

How Does the Level of CAPE Impact Estimation Performance?

With Figures 1 and 2 in mind, perhaps the superior performance of the $(1/\text{CAPE} - R)$ model is sensitive to the level of CAPE? Figure 7 shows that CAPE has ranged from 7 to 44. To evaluate the impact of the **level of CAPE** on estimation performance, we split CAPE values into three groups: low (< 15 , *i.e.*, below 25th percentile), medium (≥ 15 & ≤ 26 , *i.e.*, between 25th and 75th percentile), and high (> 26 , *i.e.*, above 75th percentile) at the time of estimation.

Figure 7: Historical Distribution of CAPE and Corresponding Values of $\ln(\text{CAPE})$ & $1/\text{CAPE}$; 4/1953 – 8/2024

Distribution	Percentile							Mean	Volatility	Min.	Max.
	1 st	5 th	25 th	50 th	75 th	95 th	99 th				
CAPE	8	9	15	20	26	36	43	21	8	7	44
Corresponding $\ln(\text{CAPE})$	2.0	2.2	2.7	3.0	3.3	3.6	3.8	3.0	0.4	1.9	3.8
Corresponding $1/\text{CAPE}$	13%	11%	7%	5%	4%	3%	2%	6%	3%	15%	2%

Source: Robert Shiller and PMA. Provided for illustrative purposes only.

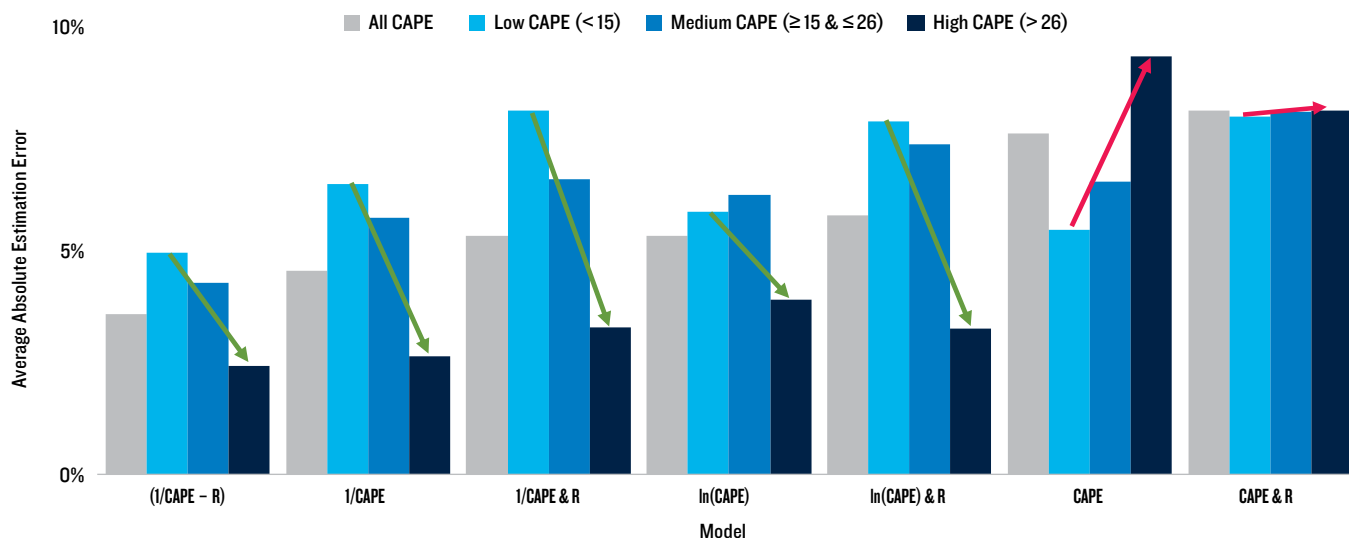
Figure 8 highlights that as the level of CAPE climbed from low to high, the estimation accuracy of both $1/\text{CAPE}$ - and $\ln(\text{CAPE})$ -based models improved while that of CAPE-based models deteriorated. We offer a potential explanation: CAPE tends to be right-skewed, with extreme values more likely associated with “market exuberance” that stretch its distribution, making predictions less reliable. In contrast, $1/\text{CAPE}$ and $\ln(\text{CAPE})$ “normalize” CAPE and scale down high CAPE values, dampening the influence of potential outliers, making the relationship with future stock returns more linear and stable, reducing estimation errors.⁹

Interestingly, $(1/\text{CAPE} - R)$ has delivered the lowest estimation error across all levels of CAPE. Given the recent high level of CAPE (35, as of August 2024, 94th percentile), $(1/\text{CAPE} - R)$ is expected to deliver a low estimation error.¹⁰

⁹ As aforementioned, theoretically, stock expected return has a linear relationship with $1/\text{CAPE}$, but not CAPE or $\ln(\text{CAPE})$.

¹⁰ We run regressions based on all CAPE levels and then – conditional on the level of CAPE (*i.e.*, all, low, medium, and high) – calculate the average estimate (of future 10y average annual stock real return) and the average upper and lower bounds of the 95% prediction interval. The average estimate of future stock returns declines as the level of CAPE increases from low to high, given their inverse relationship. However, the average width of prediction interval (*i.e.*, average upper bound – average lower bound) remains relatively stable (17%-18%) regardless of the level of CAPE, driven by the OLS assumption of constant variance of residuals (*i.e.*, homoscedasticity).

Figure 8: Average Absolute Estimation Error by the Level of CAPE (Regressions Based on All CAPE Levels); 3/1993 – 9/2014



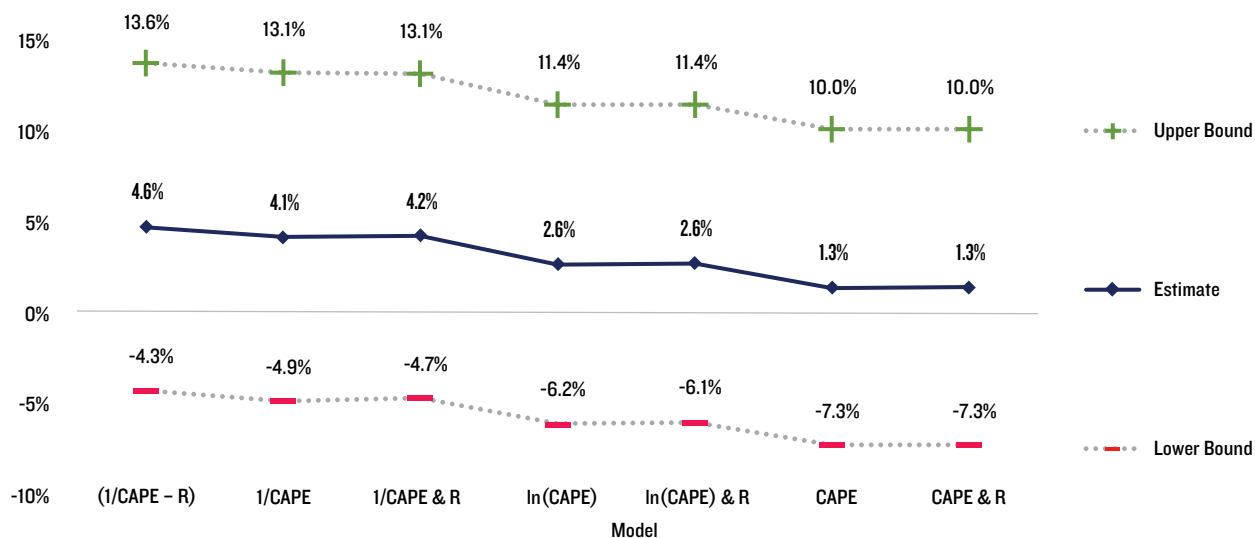
Same note and source as Figure 4. Regressions and estimations are based on the full sample with all levels of CAPE. We then calculate the average of absolute estimation error conditional on the level of CAPE. Provided for illustrative purposes only.

What Are the Market Implications for CIOs?

Recently, $1/\text{CAPE}$ has been relatively low and R has been relatively high, making $(1/\text{CAPE} - R)$ quite low. So, we should not be surprised that future stock returns are forecast to be lower than average. In addition, we recently argued that the low value of $(1/\text{CAPE} - R)$ suggests that the gap between future stock and bond returns over the next decade is estimated to be only 2.4%/y in favor of stocks *vs.* the historical average of 4.4%/y.¹¹

Based on the historical relationship, the recent reading of $(1/\text{CAPE} - R)$ corresponds to a **4.6%/y real return** over the next 10y with a 95% prediction interval ranging from -4.3%/y to 13.6%/y (Figure 9).¹² Our baseline estimate of 4.6%/y is equivalent to a **6.6%-7.6%/y nominal return** assuming 2%-3%/y inflation. In contrast, $\ln(\text{CAPE})$ – and CAPE-based models, with a weaker forecast record, generate lower **real return** estimates ranging from 1.3%/y to 2.6%/y.

Figure 9: Estimate and 95% Prediction Interval of Future 10y Average Annual Stock Real Returns; as of August 2024



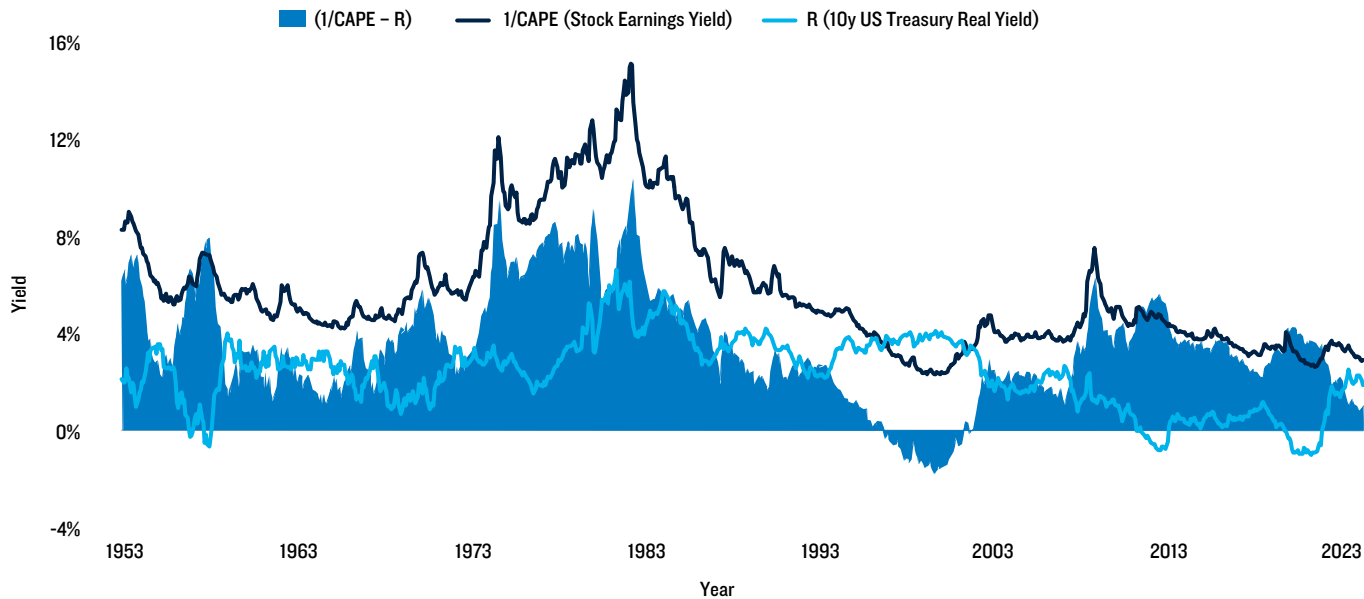
Same note and source as Figure 4. Provided for illustrative purposes only.

11 See *Higher Bond Yields & the Fed Model – Implications for Future Stock-Bond Relative Returns*, PMA, November 2023.

12 For reference, Goldman Sachs (2024) relies on $\ln(\text{CAPE})$ (and other explanatory variables) and generates a lower estimate: a 1%/y real return with a 95% prediction interval ranging from -3%/y to 5%/y (assuming 2%/y inflation). Alternatively, if we regress future 10y average annual stock real return on $(1/\text{CAPE} - R)$ *conditional on the level of CAPE being high* (i.e., > 26) given the recent level of CAPE reaching historical high, we will get a higher estimate of 4.8%/y with a narrower 95% prediction interval ranging from 0.7%/y to 8.9%/y (see Figure A3 in Appendix). However, the results are sensitive to the choice of the threshold for high CAPE and are subject to biases.

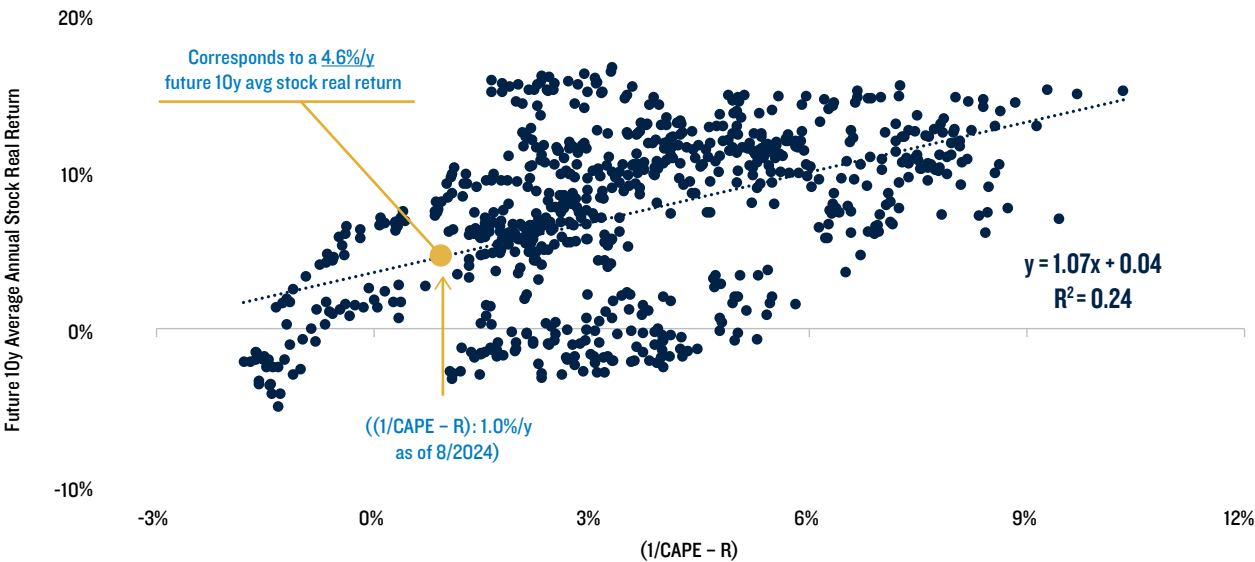
Appendix

Figure A1: 1/CAPE (Stock Earnings Yield), R (10y US Treasury Real Yield) & (1/CAPE - R); 4/1953 - 8/2024



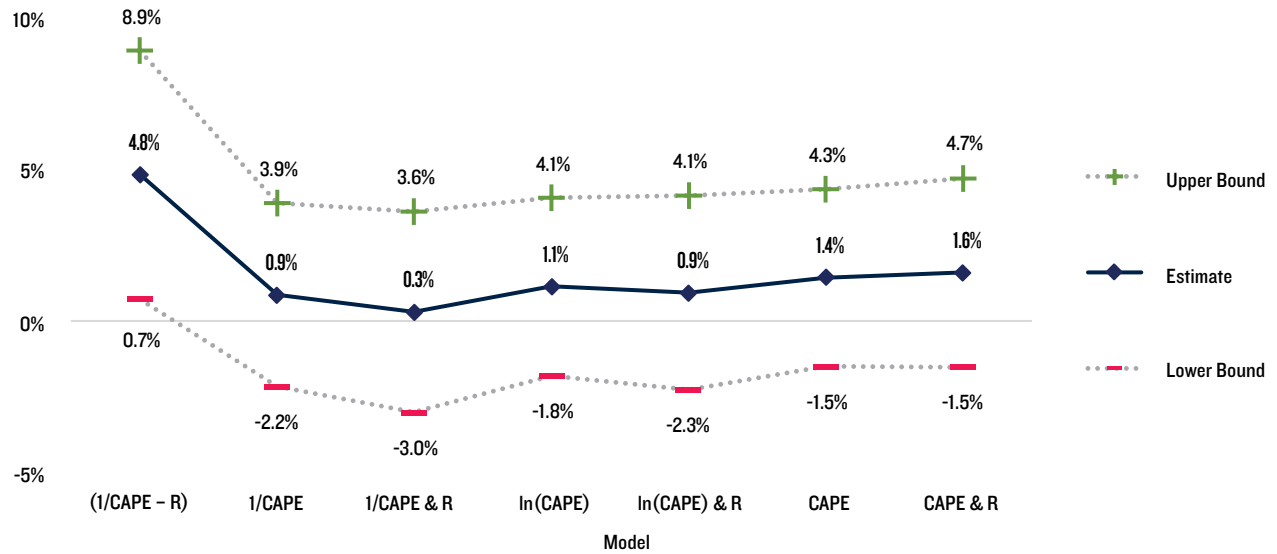
Same note and source as Figure 4. Provided for illustrative purposes only.

Figure A2: Regression of Future 10y Average Annual Stock Real Returns on (1/CAPE - R); 4/1953 - 9/2014



Same note and source as Figure 4. Provided for illustrative purposes only.

Figure A3: Estimate and 95% Prediction Interval (Regressions Conditional on the Level of CAPE Being High (*i.e.*, > 26)); as of August 2024



Same note and source as Figure 4. Provided for illustrative purposes only.

CAPE Models Specification Details

$$(1/CAPE - R) \text{ model as of year } t: \quad E(r_{t,t+10}) = \alpha_t + \beta_t \times (1/CAPE_t - R_t) + \varepsilon_{t,t+10} \quad (A1)$$

$$1/CAPE \text{ model as of year } t: \quad E(r_{t,t+10}) = \alpha_t + \beta_t \times (1/CAPE_t) + \varepsilon_{t,t+10} \quad (A2)$$

$$1/CAPE \text{ \& R model as of year } t: \quad E(r_{t,t+10}) = \alpha_t + \beta_t^{1/CAPE_t} \times (1/CAPE_t) + \beta_t^{R_t} \times R_t + \varepsilon_{t,t+10} \quad (A3)$$

$$\ln(CAPE) \text{ model as of year } t: \quad E(r_{t,t+10}) = \alpha_t + \beta_t \times \ln(CAPE_t) + \varepsilon_{t,t+10} \quad (A4)$$

$$\ln(CAPE) \text{ \& R model as of year } t: \quad E(r_{t,t+10}) = \alpha_t + \beta_t^{\ln(CAPE_t)} \times \ln(CAPE_t) + \beta_t^{R_t} \times R_t + \varepsilon_{t,t+10} \quad (A5)$$

$$CAPE \text{ model as of year } t: \quad E(r_{t,t+10}) = \alpha_t + \beta_t \times CAPE_t + \varepsilon_{t,t+10} \quad (A6)$$

$$CAPE \text{ \& R model as of year } t: \quad E(r_{t,t+10}) = \alpha_t + \beta_t^{CAPE_t} \times CAPE_t + \beta_t^{R_t} \times R_t + \varepsilon_{t,t+10} \quad (A7)$$

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